Onasvee Banarse

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Education

University of Massachusetts Amherst — Amherst, MA
Master's in Computer Science — Expected Dec 2027
- Relevant Coursework: Machine Learning, Advanced Algorithms , Distributed and Operating Systems

Savitribai Phule Pune University — Pune, India
Bachelor of Technology (B.Tech) in Computer Science **[GPA 9.38/10]** — Aug 2019 – Mar 2023
- Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Object Oriented Programming

Technical Skills

Languages: Python, Java, JavaScript, C/C++, SQL (PostgreSQL), Groovy, HTML/CSS
Frameworks & Tools: Node.js, Express.js, Selenium, JMeter, SoapUI
Developer Tools: Git, PyCharm, IntelliJ, VS Code, Docker, JMeter CLI, SoapUI Test Runner
Databases: PostgreSQL, CassandraDB
Libraries: NumPy, pandas, Matplotlib, Scikit-learn, TensorFlow

Experience

SDET / Associate QA Engineer — December 2023 – December 2025
Reliance Jio Platforms Limited — Navi Mumbai, India
- Developed Selenium automation frameworks in Java for web application testing, integrating API calls to validate backend data against UI elements and reducing regression testing effort by 10+ hours per week.
- Performed functional and load testing of backend REST APIs using JMeter, Postman, and SoapUI, achieving 85% test coverage across critical Identity and Access Management (IDAM) modules.
- Designed and executed end-to-end test scenarios for JioID SSO, validating OAuth2/OIDC authentication flows, token handling, and client-specific configurations across multiple integrations.
- Collaborated with backend engineers to debug production issues, analyze service logs, and trace data inconsistencies across relational and NoSQL systems, reducing defect leakage by 30%.
- Performed cross-database validation using JMeter (JDBC) and SQL queries to identify data inconsistencies across Cassandra clusters and PostgreSQL.

Cyber Security Intern — January 2022 – June 2022
Acmegrade Pvt Ltd — Bengaluru, India
- Conducted vulnerability assessments and penetration testing on internal web applications using OWASP methodologies to identify security weaknesses and recommend remediation strategies.
- Analyzed network traffic and security logs using Wireshark and SIEM tools to detect anomalies and potential intrusions, reducing false positives by 60% through alert tuning.
- Reverse-engineered malware samples to analyze malicious behavior and develop detection signatures for internal threat intelligence systems.

Projects

Image Manipulation and Forgery Detection | Python, TensorFlow, ResNet, Streamlit — Jul 2022 – Jul 2023
- Built and deployed a CNN-based deep learning pipeline using TensorFlow and ResNet with a Streamlit interface to detect manipulated regions in digital images, trained on CASIA v1.0 and v2.0 datasets.
- Designed preprocessing and evaluation workflows using ELA for feature extraction, handling copy-move, splicing, and retouching manipulation types; achieved 95.30% accuracy on CASIA v1.0 and 91.31% on CASIA v2.0, with results consistent with state-of-the-art deep learning approaches reviewed across 20+ prior works.

Data Migration Utilities | Node.js, Express.js, PostgreSQL, CassandraDB — Aug 2024 – Nov 2025
- Built REST-based migration services using Node.js and Express.js to migrate legacy CassandraDB records into updated schemas following table field restructuring, with structured validation, logging, and error handling.
- Ensured consistency via atomic transactions in PostgreSQL and idempotent write strategies in Cassandra; validated accuracy through pre- and post-migration checksums, with zero data loss confirmed post-deployment.
- Adopted by the backend engineering team for production migrations.

Test Automation Management Tool | Python, JMeter, SoapUI, Groovy — Jan 2025 – Nov 2025
- Developed a Python tool to automatically organize fragmented JMeter and SoapUI test suites by module and priority tags, reducing manual organization effort by 1–2 working days per release cycle.
- Implemented Groovy scripts to dynamically parse test files and generate regression-ready configurations, enabling consistent and repeatable execution across multiple test environments.

Publications

- Co-authored "Unmasking the Deception: An Image Manipulation Detection Using ResNet and UNet Architecture", IRJET, Vol. 12, Issue 03, Mar 2025.
- Co-authored "Exploring the Potential of Generative Adversarial Networks: A Comparative Study of GAN", IRJET, Vol. 10, Issue 03, May 2023.
- Co-authored "Autonomous UAV Swarms: Powering Up with AI", focused on AI applications in UAV swarm systems, IJRASET, Vol. 11, Issue 06, Jun 2023.